

Fast Implementation of Fully Iterative Scatter Corrected OSEM for HRRT Using GPU

Kyung Sang Kim and Jong Chul Ye*

Abstract—Accurate scatter correction is especially important for high-resolution 3D PETs due to the lack of inter-slice septa. To address this problem, a fully 3D iterative scatter-corrected OSEM in which a 3D single scatter simulation (SSS) is alternatively performed with a 3D OSEM reconstruction until convergence was recently proposed. However, due to the computational complexity of both SSS and OSEM algorithms for a high-resolution 3D PET, it has not been widely used in practice. The main objective of this paper is, therefore, to accelerate the fully 3D iterative scatter-corrected OSEM using a Graphics Processing Unit (GPU) and verify its performance. For SSS implementation, a sinogram-driven approach is found to be more appropriate compared to a detector-driven approach, as fast linear interpolation can be performed in sinogram domain through the use of texture memory. Furthermore, a pixel-driven backprojector and a ray-driven projector can be significantly accelerated by assigning threads to voxels and sinograms, respectively. Using Nvidia's GPU (Tesla C1060) and the Compute Unified Device Architecture (CUDA), the execution time of the SSS is less than 25(s), a single iteration of OSEM takes 25(s) for the span 9 HRRT geometry.

I. INTRODUCTION

MODERN positron emission tomography (PET) systems such as the High-Resolution Research Tomograph system (HRRT, Siemens) were designed as a fully 3D system capable of high sensitivity and enhanced resolution. In contrast to 2D PETs with inter-slice septa, coincident detection between two different rings in the 3D PET provides 3D line-of-response (LoR) data, which greatly improves the sensitivity and resolution. However, due to the lack of inter-slice septa, artifact correction has become more complicated.

To remove the scatter contributions from the measured data, several methods have been proposed for accurate scatter estimation. Monte Carlo (MC) simulation has proven to be accurate for scatter estimation [1]. MC simulation calculates the path of each photon by considering multiple scattering events. However, MC simulation is computationally very demanding for a 3D PET [2]. Hence, Watson *et al* [3] developed what they termed the single scatter simulation (SSS) algorithm. The SSS algorithm is a fast approximation of MC simulation that considers only single Compton scatter events while incorporating physical properties such as Compton effects as expressed by the Klein-Nishina formula [4]. The SSS algorithm then calculates the contribution of single scattering effects from many scatter points placed within the object, and the scatter contribution is expressed as the summation of scatter photons generated from many scatter points which are

randomly distributed within the object. In clinical applications, the SSS algorithm is now one of most popular methods for scatter correction due to its relatively high level of accuracy and reduced computational complexity. However, the computational complexity of SSS is nonetheless too high for HRRT.

To accelerate the execution speed for SSS algorithm, Watson [5] implemented a fast SSS algorithm that takes into account the cylindrical symmetry of a PET scanner. His SSS implementation includes a lookup table search with respect to the four ray sums from each scatter point to the detectors. However, the algorithm is not fully 3D because scatter events along the tilted planes are not considered. Werling *et al.* and Accorsi *et al.* extended this idea and implemented similar 3D SSS algorithms [6], [7]. Accorsi *et al.* [7] tested their implementation using the Allegro scanner (Phillips Medical Systems). Even with the relatively low resolution of the Allegro scanner as compared to the HRRT scanner, the execution time of SSS was nearly 5 min for a whole-body assessment.

II. FULLY ITERATIVE SCATTER-CORRECTED OSEM

Proposed algorithm repeats 3D-SSS and 3D-OSEM until the final reconstruction of the radiotracer does not change. More specifically, the first execution of SSS uses a scatter-uncorrected image to obtain a scatter sinogram. After reconstructing a radiotracer image from the scatter-corrected sinogram, it becomes the new input for 3D-SSS to estimate the scatter distribution. This procedure is iterated until convergence. Therefore, one of the objectives of this paper is to allow fast implementation so as to demonstrate that 3D iterative scatter correction is indeed effective on HRRT geometry. To show the effectiveness and practicality of the fully 3D iterative scatter-corrected OSEM, we perform simulations using actual experiments with HRRT. Our results demonstrate that the fully 3D iterative scatter-corrected OSEM is indeed effective in improving the resolution.

III. GPU IMPLEMENTATION OF PROJECTION, BACKPROJECTION AND SINOGRAM-DRIVEN SSS

First, we define geometry parameters of Δx , Δy , Δz , $\cos \phi$, $\sin \phi$, and D using constant memory, as all kernels can access constant memory as rapidly as they can. Now, Figs. 1(a) and (b) show descriptions of a projection and a backprojection optimized for a GPU, respectively. More specifically, when calculating a projection, a thread is assigned to a sinogram pixel, and each run concurrently with a ray-driven projector. Here, an object is bound to a texture memory. Note that a texture memory is useful for fast linear interpolations. While a

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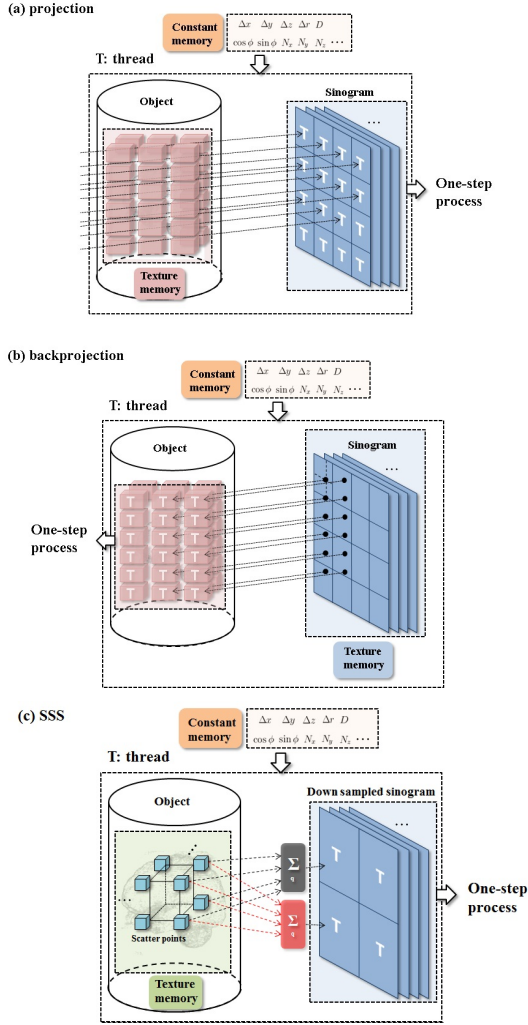


Fig. 1. Assignment of a constant memory, threads and a texture memory of GPU for (a) a projection, (b) a backprojection, and (c) sinogram-driven SSS.

linear interpolation procedure is a computationally demanding process in a CPU, a GPU supports fast 1D, 2D and 3D linear interpolation with floating coordinates using texture memory. Hence, this is very useful for calculating a line integral in a projection. Furthermore, during a projection, the values of the objects do not vary. Hence, binding the object to read-only texture memory is not a problem. Thus, the texture memory is bound to the object to allow rapid calculation of the projection. In a backprojection, a sinogram is bound to texture memory, and a thread is assigned to an object voxel. Here, each backprojection can be calculated concurrently using a pixel-driven method as shown in Fig. 1(b). As the pixel-driven backprojector requires linear interpolation in the sinogram domain, interpolation using texture memory is fast.

IV. RESULTS

A. Reconstruction quality evaluation

We evaluated the proposed algorithm using a uniform phantom experiment. The uniform phantom in Fig. 2 is a cylindrical phantom, in which the radiotracers are distributed uniformly. To verify the performance, we compared the uniformity ratio. Here, the uniform phantom is divided by inner and outer parts, where the inner part has an area with a radius of $1/8 \sim 2/8$ and the outer part has a radius of $6/8 \sim 7/8$. We then calculated the averages of the inner and the outer parts; the uniformity ratio is defined as the ratio between radius inner and outer averages. Due to the uniform distribution of radiotracers, the contrast should be 1. However, as shown in Table I, a scatter-uncorrected image reconstruction indicates that the average of radius inner part is much higher than that of radius outer part due to the dominant scattering at the center. We also observed that the uniformity ratio converges to nearly 1 after 3 or 4 iterations of the proposed method. This indicates that the fully 3D iterative scatter-corrected OSEM algorithm provides a more accurately scatter-corrected reconstruction image.

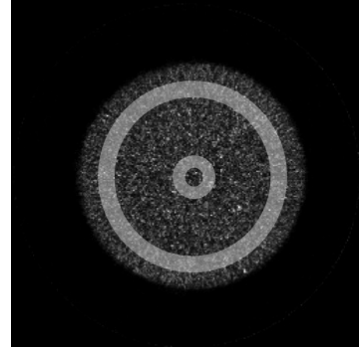


Fig. 2. Reconstruction of a uniform phantom and the location of inner and outer rings to calculate the uniformity ratio.

Iterations	Inner average	Outer average	Ratio (inner/outer)
0	0.2924	0.1904	1.5378
1	0.0925	0.1098	0.8422
2	0.1224	0.1175	1.0418
3	0.1133	0.1153	0.9825
4	0.1158	0.1159	0.9985

TABLE I
UNIFORMITY RATIO AT EACH ITERATION.

B. Execution time for HRRT

The algorithm was implemented on an Nvidia Tesla C1060 system, and we used span9 sinogram data with a dimension of $256 \times 288 \times 2209$ including 15 tilted planes, and object with a dimension of $256 \times 256 \times 207$. The calculation time is summarized in Table. II. Significant acceleration was observed.

STEP	GPU (s)	CPU (s)	Acc.
SSS	25	80 (2D-SSS)	3.2x
Pixel-driven backprojector	5.8	1100	190x
Ray-driven projector	14.8	900	60.8x
OSEM(1 iter.)	25	2000	80x

TABLE II
EXECUTION TIME FOR HRRT USING NVIDIA TESLA C1060, AND INTEL
I7 3.07GHZ 4GB.

V. CONCLUSIONS

We derived a fully 3D iterative scatter-corrected OSEM and described a GPU implementation of this for high-resolution research tomograph (HRRT). For maximal acceleration of the GPU implementation, we found that a ray-driven projector, a pixel-driven backprojector and a sinogram-driven SSS algorithm were appropriate. We demonstrated that the proposed GPU implementation allows fast and accurate fully 3D iterative scatter-corrected OSEM results.

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